MILITARY SPECIFICATION

MASK, GAS, ROCKET PROPELLANT, M26A1

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of mask and replacement canister that protects the respiratory system against low concentrations of toxic vapors and aerosols incident to the handling of rocket propellants.

2. APPLICABLE DOCUMENTS.

2.1 **Government documents.** The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-I-559 — Ink, Marking Stencil, Opaque, for Porous Surfaces (Wood, Boxes, Fiber Cartons, etc.).

CCC-C-419 — Cloth, Duck, Cotton, Unbleached, Plied-Yarns, Army and Numbered.

CCC-D-950 — Dyeing and After Treating Processes for Cotton Cloths.

PPP-B-636 — Box, Fiberboard.

MILITARY

MIL-P-116 — Preservation, Methods of.

MIL-W-530 — Webbing and Tape, Textile, Cotton, General Purpose, Natural or in Colors.

MIL-P-7254 — Propellant, Nitric Acid.

MIL-D-8634 — Decal, Elastomeric Pigmented Film, For Use on Exterior Surfaces.

STANDARDS

FEDERAL

FED-STD-595 — Colors.

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

FSC 4240

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MIL–STD–129 — Marking for Shipment and Storage.

DRAWINGS

EDGEWOOD ARSENAL

C5–85–15 — Hose.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules, Ratings and Regulations.

(Application for copies of these freight classification rules, ratings and regulations should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois.)

3. REQUIREMENTS

3.1 Components.

3.1.1 *Canister (see 6.4).* The canister shall meet the performance requirements of table I. The body shall be metallic and its height shall be between 7⅞ and 10 5/16 inches. Its girth shall be between 15¼ and 17½ inches. The canister neck shall be made of brass or aluminum and shall be threaded in order to mate with the male coupling of the hose. The canister shall be equipped with a nitric acid resistant particulate filter.

3.1.1.1 *Window indicator.* Each canister shall be equipped with a window indicator which shall give a visual indication of approaching exhaustion of the canister. Visual indication shall be apparent when no less than 50 percent nor more than 75 percent of the canister service life has been exhausted when tested as specified in 4.6.3 and 4.6.4. An instruction label which provides proper interpretation of window indications shall be placed near the window indicator.

TABLE I. *Canister performance requirements*

Characteristic	Requirement	Test paragraph
Airflow resistance, maximum	65 millimeters	4.6.1
Airflow resistance, (after equilibration, at 80% relative humidity) maximum.	75 millimeters	4.6.1
DOP smoke penetration (before gas life tests) maximum.	0.015 percent	4.6.2
Gas life, red fuming nitric acid, minimum.	20 minutes at an intermittent flow rate of 25 liters per minute and a concentration of 1% acid at 50% relative humidity.	4.6.3
Gas life, unsymmetrical dimethyl hydrazine (UDMH), minimum.	5 minutes at an intermittent flow rate of 25 liters per minute and a concentration of 2% UDMH at 50% relative humidity.	4.6.4

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TABLE I. *Canister performance requirement*—Continued

Characteristic	Requirement	Test paragraph
Gas life, carbon tetrachloride, minimum.	12 minutes at a constant flow rate of 64 liters per minute and a concentration of 2% carbon tetrachloride.	4.6.5

3.1.1.2 *Canister marking.* Each canister shall be painted orange no. 12246 with a ½-inch stripe, grey no. 36280, at the top. Colors shall be as specified in FED–STD–595. As an alternative to the painted stripe a ½-inch wide adhesive tape of the required color may be used. The canister shall be labeled with a decal conforming to MIL–D–8634 or with a printed waterproof label. Alternatively, printing may be directly on the canister body. Marking shall be as follows:

"CANISTER, ROCKET PROPELLANT
GAS MASK, M15A1

Date of manufacture (month and year)

Name and address of the manufacturer

This canister is designed to protect against the following:

Hydrazine

Hydrogen peroxide

Red fuming nitric acid

Unsymmetrical dimethyl hydrazine

Organic vapors such as kerosene, aniline, and alcohol."

3.1.1.3 *Canister opening.* The bottom opening of the canister shall be covered with moisture vaporproof adhesive tape. Vaporproof plugs or caps shall be affixed to the nozzles of replacement canisters.

3.1.2 *Carrier.* A carrier of suitable commercial design with adjustable shoulder strap and adjustable waist strap shall be provided for storage and for carrying the canister, hose, and facepiece. It shall have a permanent opening in the bottom so that the adhesive tape can be removed from the canister without removing the canister from the carrier. It shall have a quick opening closure to permit rapid donning of the facepiece from the carry position. Strapping shall be provided within the carrier to hold the canister in place when the facepiece is withdrawn for donning.

Appropriate padding shall be provided in the facepiece storage area to protect the lens. The carrier shall have a window section in the canister area to provide viewing of the window indicator and the instruction label.

3.1.2.1 *Carrier material.* The carrier body shall be made from cotton duck, type I, no. 10 hard texture, color OD–7 conforming to CCC–C–419 and shall be finished in accordance with type I, class B of CCC–D–950.

3.1.2.2 *Shoulder strap.* The shoulder strap shall be made from cotton webbing, 1½ inches wide, type II or IIa, color OD–7, class 7, conforming to MIL–W–530.

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3.1.2.3 *Waist strap.* The waist strap shall be made from cotton webbing, 1 inch wide, type II or IIa, color OD–7, class 7, conforming to MIL–W–530.

3.1.2.4 *Marking.* The carrier shall have the following marking, using stencil ink, color black 37038, conforming to FL 1–559:

"MASK, GAS, ROCKET PROPELLANT, M26A1
FOR US AIR FORCE

CAUTION: REMOVE TAPE FROM BOTTOM OPENING OF CANISTER BEFORE USING MASK."

3.1.2.5 *Prototype.* A prototype of the carrier, as designed by the contractor, shall be submitted through the Contracting Officer to the Director of Engineering and Industrial Services, U.S. Army Edgewood Arsenal, Edgewood Arsenal, Maryland, for approval of the design.

3.1.3 *Hose.* The hose shall conform to Drawing C5–85–15. The hose shall be molded from synthetic rubber or other suitable nonmetallic material. The hose shall be flexible and shall not crack when tested as specified in 4.6.7.

3.1.4 *Facepiece.* The facepiece shall be a commercial type full facepiece with monocular lens. The facepiece will be accepted on a certificate by the contractor that it is a production model and a duplicate of a facepiece from a mask which has been approved by the Bureau of Mines. When assembled to the hose and canister, leakage of the assembly shall not exceed 0.003 percent DOP-smoke penetration when tested as specified in 4.6.8.

3.2 Preproduction sample. When specified in the contract or order, a preproduction sample of M26A1 masks shall be produced in accordance with this specification for examination and tests (see 6.2).

3.3 Workmanship. The mask shall be free from foreign matter (dirt, grease, or oil) and damage (dents, scratches, or punctures).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Special provisions.

4.2.1 *Alternative inspection.* The supplier may utilize any alternative inspection procedure which will insure equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Objective evidence.* The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of sections 3 and 5 for which specific

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inspection has not been provided in this specification have been satisfied.

4.2.3 *Inspection substations.* If the supplier elects to establish inspection substations for the purpose of conducting inspection for certain listed defects, the Acceptable Quality Levels (AQL's) for each inspection substation shall be equal to the AQL which is the next numerically lower value than the AQL specified for the total listed defects. The requirements of MIL–STD–105 are applicable to each substation and to the end item.

4.3 Lotting. A lot shall consist of the masks produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions. However, no more than one lot of canisters shall be represented in any one lot of finished masks.

4.4 Sampling.

4.4.1 *For examination and nondestructive tests.* Sampling shall be conducted in accordance with MIL–STD–105. For major B defects, level S–1 shall be used.

4.4.2 *For destructive tests.*

4.4.2.1 *Gas life.* Five canisters shall be selected at random from each lot for each test.

4.4.2.2 *Nitric acid resistance of particulate filter.* Two canisters shall be selected at random from each lot for test.

4.4.2.3 *For window indicator.* Sample canisters used in gas life testing will be used for this test.

4.5 Inspection provisions.

4.5.1 *Examination and nondestructive tests.* Sample masks shall be examined and tested in accordance with the classification of defects and MIL–STD–105.

4.5.2 *For destructive tests.*

4.5.2.1 *Gas life.* Sample canisters shall be tested in accordance with 4.6.3, 4.6.4, and 4.6.5. The acceptance number is zero. Tests shall be continued until gas life is exhausted to facilitate test for the window indicator operation (4.5.2.3).

4.5.2.2 *Nitric acid resistance of particulate filter.* Sample canisters shall be tested in accordance with 4.6.6. The acceptance number is zero.

4.5.2.3 *Window indicator.* The window indicator of the canisters used for gas life tests shall be observed for compliance with 3.1.1.1. The acceptance number is zero.

4.5.3 *Classification of defects.*

4.5.3.1 *Mask, Gas, Rocket Propellant, M26A1.*

Categories	*Defects*	*Acceptance standards*
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Component missing	
102	Component incorrectly assembled	
103	Marking incorrect	
104	Workmanship (3.3)	
105	Canister dimensions incorrect	

Categories	*Defects*	*Acceptance standards*
Major A:	AQL 0.65 percent defective	
151	Resistance to airflow	4.6.1
152	DOP smoke penetration	4.6.2
153	Mask assembly leakage	4.6.8

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Categories	Defects	Acceptance standards
Major B	AQL 2.5 percent defective	
	[illegible] flexibility	4.6.7
Minor	AQL 2.5 percent defective	
201	Marking missing	
202	Protective finish missing	
203	Bottom opening of canister not taped	

4.6 Tests. Distilled water and analytical reagent grade chemicals shall be used throughout the tests. Blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.6.1 *Airflow resistance.* The resistance to airflow shall be determined using the Q112 Resistance Indicator.

4.6.2 *DOP smoke penetration.* The DOP smoke penetration shall be determined using the Q127 DOP Filter Testing Penetrometer.

4.6.3 *Gas life, red fuming nitric acid and indicator performance.* The canister shall be equilibrated at the specified relative humidity and then tested against red fuming nitric acid using the specified concentration and flow rate at 25 cycles per minute. A sample of the effluent from the canister shall be taken and analyzed for oxides of nitrogen using the ion chamber method or the following wet method: Allow ½ liter per minute of the effluent to flow through a bubbler containing 30 milliliters (ml) of concentrated sulfuric acid and then through a second bubbler containing 25 ml of 5 percent potassium permanganate solution and then through a third bubbler containing 25 ml of 10 percent potassium iodide solution, 5 ml of 0.1 percent starch solution and 5 x 10^{-4} milliequivalents of sodium thiosulfate. When the third bubbler changes to a bluish-brown color, replace it with a fresh bubbler. Replace the third bubbler until the color change occurs in one minute or less. This constitutes a leakage of 5 parts per million (ppm) of NO_2. If the leakage concentration is in excess of 5 ppm within 20 minutes, the canister does not meet the requirement. The window indication of the canister shall be apparent when no less than ½ nor more than ¾ of the canister service life to a 5 ppm NO_2 leakage has been exhausted.

4.6.4 *Gas life, UDMH and indicator performance.* The canister shall be equilibrated to the specified relative humidity and then tested against UDMH using the specified concentration and flow rate at 25 cycles per minute. A sample of the effluent from the canister shall be taken continuously and analyzed for UDMH using the ion chamber method or the following wet method: Pass one liter per minute of the effluent through a bubbler containing 15 ml of 8.3 x 10^{-5} N sulfuric acid to which has been added 2 or 3 drops of methyl red indicator solution. When the bubbler changes from pink to orange or yellow color, replace it with a fresh bubbler. Replace the bubbler until a color change occurs in one minute or less. This constitutes a leakage of 0.5 ppm UDMH. If the leakage concentration is in excess of 0.5 ppm within 5 minutes, the canister does not meet the requirement. The window indication of the canister shall be apparent when no less than ½ nor more than ¾ of the canister service life to a 0.5 ppm UDMH leakage has been exhausted.

4.6.5 *Gas life, carbon tetrachloride.* The CCl_4 gas life shall be determined in accordance with method used by Bureau of Mines which given in Appendix.

4.6.6 *Nitric acid resistance of the particulate filter.* The particulate filter shall be removed from the canister and 10 to 15 drops of nitric acid conforming to type

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III of MIL–P–7254 shall be placed on the filter. If there is visible decomposition of the filter after 10 minutes, the filter is unacceptable. A slight discoloration does not indicate decomposition.

4.6.7 *Hose flexibility (Dwg C5–85–15).* The hose shall be exposed to a temperature of −40° ± 1°F for 4 hours. While at that temperature, the hose shall be bent 360° around a mandrel that has a diameter of 4 inches.

4.6.8 *Mask assembly leakage.* The mask assembly leakage shall be determined using the Q14 Gas Mask Leakage Tester.

5. PREPARATION FOR DELIVERY

(For repair parts packaging, see 6.3.)

5.1 Packaging. Level shall be as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Mask assembly.* No preservative compounds shall be applied to any portion of the mask assembly. The mask assembly shall be placed inside the carrier provided and packaged in accordance with method 1A–15 of MIL–P–116. The inner container shall be a style RSC, type CF, class domestic, variety SW, fiberboard box conforming to PPP–B–636.

5.1.1.2 *Replacement canisters.* Each canister shall be packaged in accordance with method 1A–15 of MIL–P–116. The inner container shall be a style RSC, type CF, class domestic, variety SW, fiberboard box conforming to PPP–B–636.

5.1.2 *Level C.*

5.1.2.1 *Mask assembly.* The mask assembly, placed inside the carrier, shall be packaged to provide adequate protection against corrosion, deterioration, and damage from the supply source to the first receiving activity for immediate use.

5.2 Packing. Level shall be as specified (see 6.2).

5.2.1 *Level A.*

5.2.1.1 *Mask assembly.* Ten mask assemblies packaged as specified in 5.1.1.1 shall be packed in a style RSC, V3c box conforming to PPP–B–636. The container shall be closed in accordance with the requirements for weather-resistant, style RSC box in the appendix thereto. The packed and closed box shall be reinforced with zinc-coated (galvanized) metal banding in accordance with the appendix to PPP–B–636.

5.2.1.2 *Replacement canisters.* Ten canisters, packaged as specified in 5.1.1.2, shall be packed in a style RSC, V3c, fiberboard box conforming to PPP–B–636. The container shall be closed in accordance with the requirements for weather-resistant, style RSC boxes in the appendix thereto. The packed and closed box shall be reinforced with zinc coated (galvanized) metal banding in accordance with the appendix to PPP–B–636.

5.2.2 *Level C.*

5.2.2.1 *Mask assembly.* Mask assemblies packaged as specified in 5.1.1.1 shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in accordance with Uniform Freight Classification rules, ratings, or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, packages and shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

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6.1 Intended use. The mask covered by this specification is intended for protection of the eyes and respiratory system against rocket propellant fumes (see 3.1.1.2).

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Items to be procured.

(c) Level of packaging and packing required.

(d) Whether a preproduction sample is required.

(1) Quantity required.

(2) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.

(3) Name and address of test facility and shipping instructions when testing is performed by the Government.

(4) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Repair parts packaging. When repair parts packaging is required for items authorized for stocking as repair parts, the repair parts packaging data sheets covering the applicable components of the M26A1 mask should be used.

(Copies of repair parts packaging data sheets may be obtained from the Director of Engineering & Industrial Services, ATTN: Chief, Design Branch, US Army Edgewood Arsenal, Edgewood Arsenal, Maryland 21010.)

6.4 Canister filling. The following canister filling information is advisory only. Whether or not this information is used, the canister must meet the requirements of this specification. Edgewood Arsenal has found that the following filling, from bottom to top, has made a satisfactory canister:

(a) A fuming nitric acid resistant particulate filter.

(b) 400 cc of 8 to 14 mesh soda lime conforming to MIL–S–17063 except that it does not contain an indicator and does not have five percent moisture.

(c) 300 cc of 6 to 16 mesh silica gel manufactured by Davison Chemical Division of W. R. Grace and Company.

(d) 800 cc of 8 to 14 mesh, copper impregnated, activated charcoal.

Custodians:

Army—MU

Air Force—84

Review interest:

Army—MD

Air Force—84

Preparing activity:

Army—MU(EA)

Project No. 4240-0148

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APPENDIX

BUREAU OF MINES CANISTER TESTING APPARATUS

10. Figure 1 shows the arrangement of the various parts of the Bureau of Mines canister testing apparatus. It may be used for testing canisters against both gases and vapors, but only that portion which is used for testing against vapors is described in detail in this section. Items 6 to 27 on the drawing are not used for this purpose and may be disregarded.

10.1 The 20-liter bottles (2) of the humidifying train are about one-third full of a sulfuric acid-water mixture adjusted to a specific gravity of 1.364 (43.13 percent H_2SO_4) so that the relative humidity of the effluent air is 50 percent. Some slight adjustment of this solution may be necessary to obtain the proper relative humidity of the effluent air. Bottle 3 contains loosely packed glass wool to stop spray from the other bottles.

10.2 Details of the constant airflow regulator (5) are given in the following reference: Oberfell, G. C. and Mase, R. P., An Automatic Compensating Flowmeter: Ind. Eng. Chem. Vol. 11, April 1919, pp. 294–6.

10.3 The parts numbered 28 to 33, inclusive, comprise the carbon tetrachloride displacer and vaporizer. A more detailed sketch of this assembly is shown in figure 2. A self-starting, clock-type synchronous motor (A), with reducing gears that provide a shaft speed of one rpm is connected to a worm-drive reducing gear (B), which has a ratio of 100 to 1. A flat-grooved plastic pulley (C), is held in place on the shaft of the worm gear by the friction burr (D). One end of a piece of no. 26 B & S gage chromel wire (E), is inserted through a small hole in the bottom of the pulley groove which connects with a ⅛-inch hole at right angles to it.

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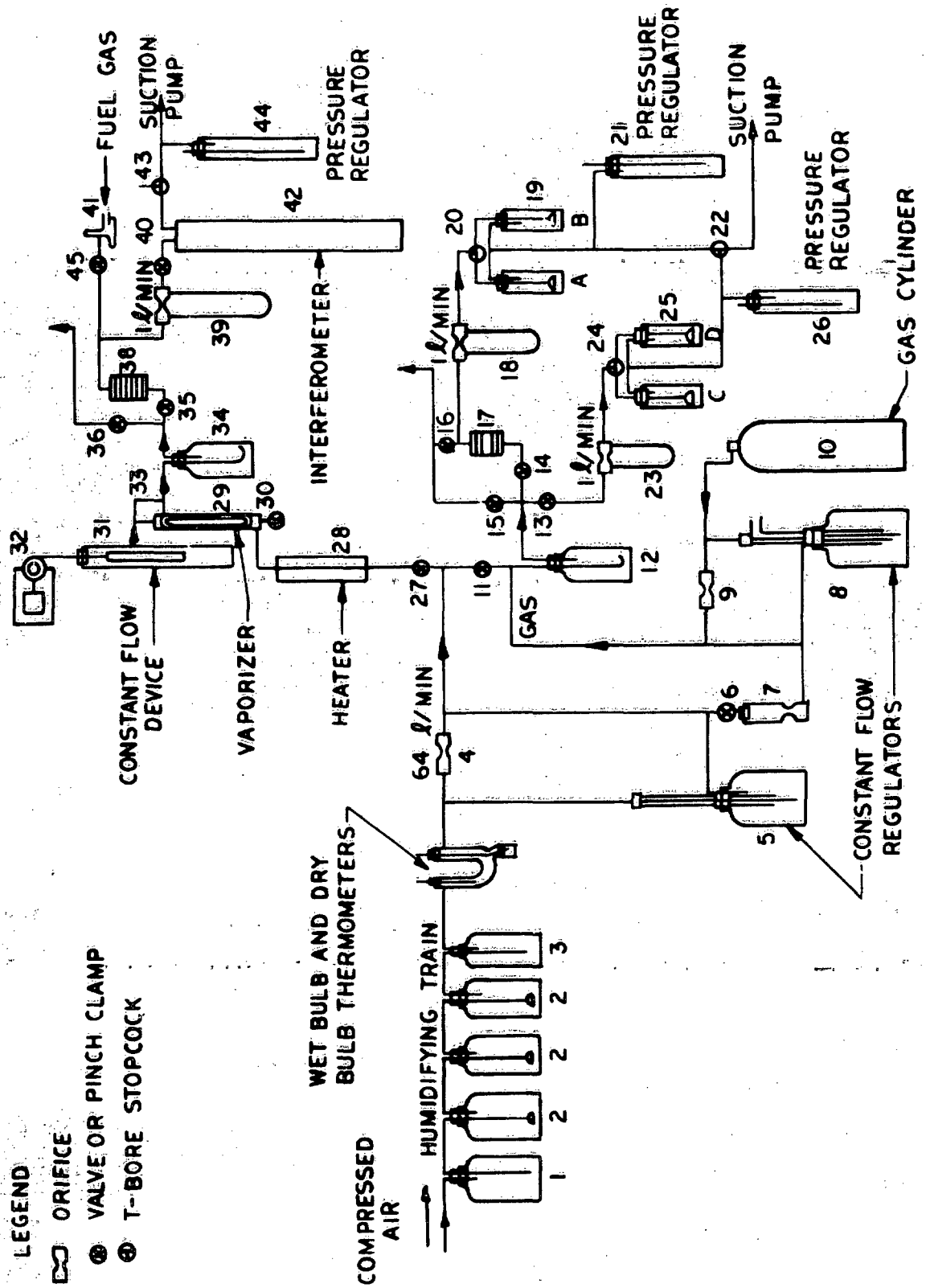
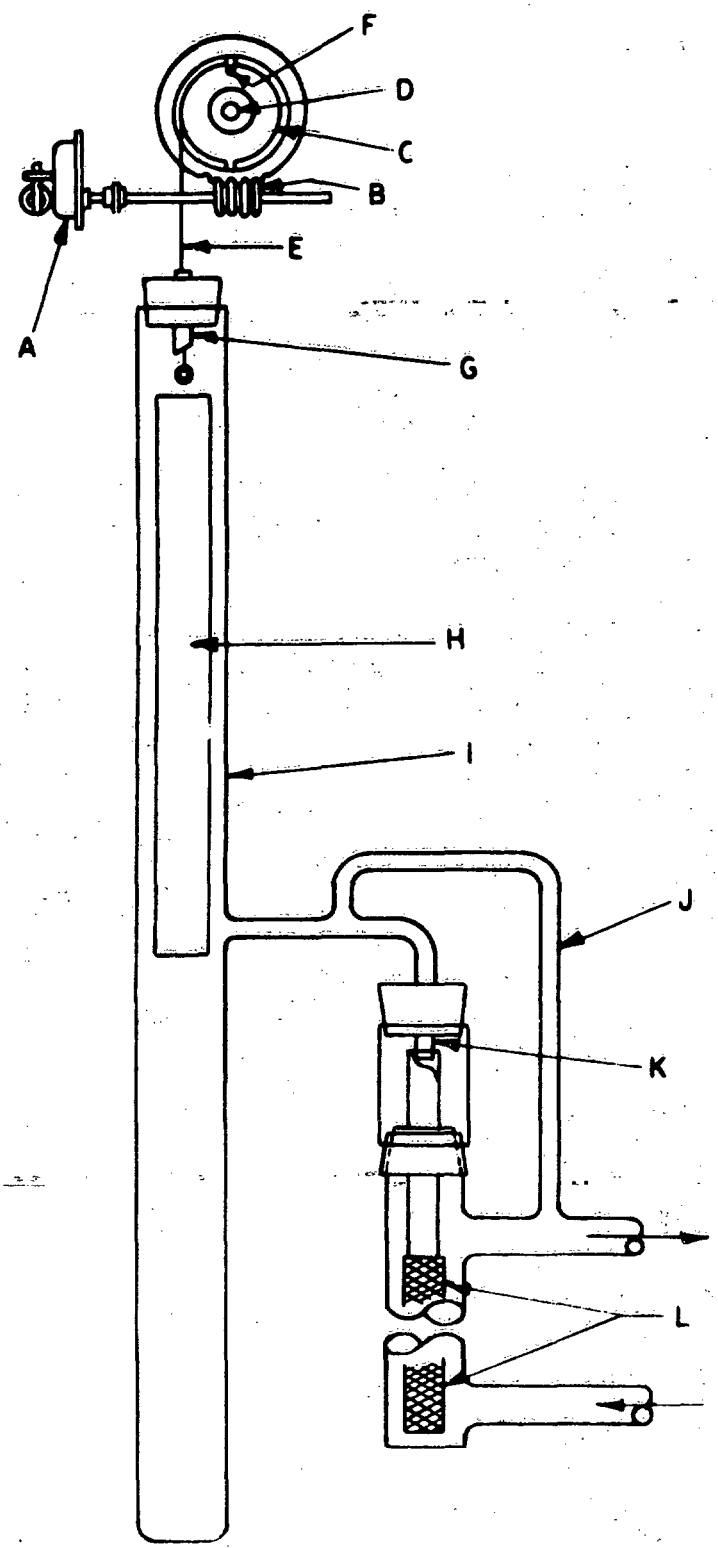


Figure 1
Sketch of apparatus for testing canisters against gases or vapors

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CCl_4 Displacer and vaporizer

Figure 2

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The wire is secured by a tapered plug (F), which is forced into the 1/8-inch hole. The wire is wound around the pulley enough turns to equal the distance to be travelled by the stainless steel plunger (H). The wire is fed through a short section of capillary glass tubing in the rubber stopper which closes the well (I), then through a small hole (pierced by a hot wire) in the rubber policemen (G), fitted to the bottom of the capillary tubing. The ends of the capillary tubing should be fire-polished to provide a smooth bearing surface for the wire. The policeman serves as a seal to prevent outward leakage of air. The wire it attached to the plunger (H). The diameter of the plunger and the diameter of the plastic wheel (C), at the bottom of the flat groove are calculated as follows:

Dimensions and calculations for liquid displacer
(One revolution of the pulley in 100 minutes)

For 1000 ppm. CCl_4 at 32 liters/minute

Flat drum pulley = 4.07 cm diameter

With 26 gage chromel wire

Displacer = 7/16 inch diameter = 1.111 cm

For 1000 ppm. CCl_4 at 64 liters/minute

Above pulley and wire

Displacer = 5/8 inch diameter = 0.625 inch

Displacer calculated from above should be 0.618 inch or 1.570 cm diameter.

The difference is about 1 percent.

For calibration, the delivery rate of liquid CCl_4 shall be measured and concentration in ppm at standard conditions of 25°C and 760 mm Hg. shall be calculated.

For example, if 10.0 cc of liquid CCl_4 are delivered in 77.5 minutes

1.584 = sp gr of CCl_4 at 25°C

153.83 = molecular weight

24.45 liters = gram molecular volume at 25°C and 760 mm Hg.

Then for an airflow of 32 liters/minute

$$\frac{10.0 \times 1.584 \times 24.45}{153.83 \times 77.5} = 0.0325 \text{ liters } CCl_4 \text{ vapor/minute.}$$

$$\frac{0.0325 \times 1{,}000{,}000}{32.03} = 1015 \text{ ppm } CCl_4 \text{ in the flowing mixture.}$$

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The liquid is displaced at a constant rate through the side-arm of the well and runs down a coiled section of chromel wire (K) to a wick (L) in the vaporizing tube. The wick is a stocking of muslin drawn over a section of glass tubing which is sealed at both ends. A pressure-equalizing tube (J) serves to equalize the pressures above the liquid in the well and in the vaporizer, and allows a smooth flow of the liquid onto the cloth wick in the vaporizer. The heater, no. 28 in figure 1, is a tube furnace, the temperature of which is adjusted so that the liquid wets no more than half of the wick and maintains a constant rate of evaporation. For the concentration of carbon tetrachloride vapor (1000 ppm) used in testing chemical cartridges, the heater will probably not be needed.

Dimensions for liquid displacer
(One revolution of the pulley in 100 minutes)

For 5,000 ppm CCl_4 at 32 liters/minute

Flat drum pulley = 4.07 cm diameter

With 26 gage chromel wire

Displacer = 2.60 cm diameter

Weight of displacer = 1333 grams

Calibrated 5550 ppm.

For 20,000 ppm CCl_4 at 64 liters/minute

Flat drum pulley = 14.29 cm diameter with 26 gage chromel wire

Displacer = 3.84 cm diameter

Weight of displacer = 1898 grams

Calibrated 20,900 ppm

The length of the displacers is 20¾ inches

10.4 For 1,000 ppm. of carbon tetrachloride vapor, the ouside diameter of the well (I) is 1⅛ inches, for 5,000 and 20,000 ppm. of carbon tetrachloride vapor it is 2⅛ inches. The length of the plunger (H) is about 21 inches and that of the well should be at least twice this. The length of the wick is about 30 inches. The rest of the apparatus, with the exception of the sampling lines, should be made of glass tubing of about one-half inch inside diameter. All joints should be glass-to-glass and covered with rubber tubing. The tube leading to the burner and the side-arm and equalizing tube of the dropping device should be about 4 millimeters inside diameter.

10.5 Details of the construction and method of using the Beilstein flame tester for determining the end of the life of the canister are given on separate sheets which have a drawing of the sleeve attached.

10.6 Other numbered parts on figure 1 are as follows:

4, 9, 18, 23, 29. Flowmeters

12, 34. Gas-mixing bottles

17, 38. Canisters being tested

19, 25. Petticoat-type sampling bubblers

33. Pressure compensator

36. By-pass which is used while the test mixture is being prepared and analyzed.

10.7 When the apparatus has been assembled, the test procedure is as follows: The flowmeter shall be calibrated for an air-

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flow of 64 liters per minute allowing the air to flow through by-pass (36). Valves 35 and 37 should be tightly closed to avoid contaminating the sampling line. The relative humidity of the air should be checked by means of a wet and dry bulb thermometer arrangement and adjusted, if necessary, to 50–5 percent relative humidity. The sampling tube should be disconnected from the burner at some point to the left of screw clamp (45) and the burner should be lit. The burner should burn with a yellow flame. The constant flow device, which has been previously calibrated to deliver the theoretical volume of liquid carbon tetrachloride to give 1,000 ppm. at the rate of airflow being used for this test, should be started. If needed, the heater should be adjusted as previously described. About 10 minutes shall be allowed for the apparatus to come to equilibrium. In the meantime, the cartridge or cartridges to be tested shall be weighed and their resistance to air flowing at a rate of 85 liters per minute should be measured. The cartridge or cartridges may be attached to the respirator mounted as in dust respirator tests with the exhalation valve sealed off and placed in a box so that the test mixture is forced through them in the proper direction, or a special holder may be used to hold the cartridge. Either of these arrangements should be put in the position of the canister (38 in the sketch). When the apparatus has come to equilibrium, valves 35 and 37 should be opened, valve 36 should be closed and the time should be closed and the time should be noted. After 15 to 30 seconds, the sampling tube should be reconnected to the burner and the flow of air should be adjusted by means of the screw clamp (45) so that the burner flame is blue. Once this flow has been adjusted, the screw clamp should be left in the same position and only minor adjustments need be made on subsequent tests. The test should be continued until the appearance of a light green coloration of the flame and the time should be noted. (The sensitivity of the flame test is increased if the room is darkened.) Valve 36 should be opened, valves 35 and 37 should be closed. The cartridges should be weighed, their resistance to airflow measured.

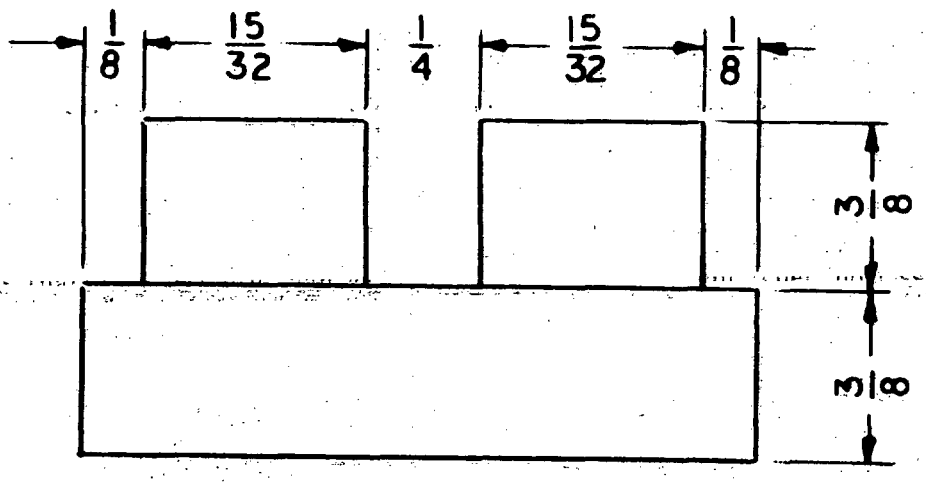
BEILSTEIN FLAME TEST FOR CARBON TETRACHLORIDE VAPORS

20.0 The following is a description of a flame test for carbon tetrachloride vapors, and a description of its use in testing the effluent from charcoal-filled chemical cartridges being tested against carbon tetrachloride capors:

20.1 The copper strip shown in figure 3 is bent in a circle to make a tube with two ¼-inch slots ⅜-inch deep. This is formed on a 27/64-inch diameter rod with the aid of a hand vise and light tapping with a hammer and then removing the burrs with a file.

20.2 The copper sleeve is attached to the top of a "Pittsburgh" burner (see Fisher Scientific Company catalog) so that bottom of the slots is even with the top of the burner tube. The burner tube and air regulator are made of brass. The burner has a flat antiflashback screen inserted flush with the top. This is made from a ⅝-inch diameter piece of steel screen of 24 mesh and about 0.015 inch diameter wire. It is formed into a cylindrical cup with sides about ⅛-inch deep by placing concentrically on a ⅜-inch id washer and forcing it through with a 5/16-inch diameter flat-ended rod.

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MATERIAL: 20 OZ. SOFT COPPER SHEET

Sleeve for CCl_4 Vapor Tester

Figure 3

20.3 The burner tube has a side inlet tube connecting at right angles so that the top edge is 1½ inches from the top of the burner. This side inlet consists of a 2-inch length of ¼-inch od thin wall brass tubing threaded into a tapped hole in the burner tube.

20.4 The burner is operated with its primary air inlet closed with adhesive tape. The primary air for the flame is the sample air through the side inlet. It is conducted to the side inlet through about ¼-inch id shell glass tubing. The glass tubing is connected end-to-end to the side inlet tube with rubber tubing. A screw clamp on a short section of rubber tubing in the sampling line about a foot away from the burner controls the sample air flow.

20.5 For use with Pittsburgh natural gas (about 1100 BTU/cu ft) the gas flow is adjusted to give a flame 7/16-inch above

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the copper indicator without airflow through the side inlet tube. A small tip of luminous flame occurs without the airflow. When coke oven gas (530 BTU/cu ft) was used, it was necessary to adjust the gas flow to give a flame 5/8-inch above the sleeve to give the same sensitivity.

20.6 The airflow to the cartridge test apparatus is then started with a charcoal cartridge or cartridges, similar to those to be tested, in place to remove any possible traces of carbon tetrachloride vapors.

20.7 The airflow to the side inlet tube of the test burner is adjusted by gradually opening the screw clamp on the sampling line. As the airflow is increased, the flame becomes larger and luminous throughout, then reduces in size and becomes nonluminous; an inner cone develops which shortens with increasing airflow and breaks when its apex nears the top of the copper sleeve; then with further increase of airflow, a cone reappears with reverse contour descending into the slot. The sample flow is finally adjusted so that the apex of this cone appears to be half way down in the slot. This cone may be irregular at first until the burner and copper sleeve heat up.

20.8 The cartridge or set of cartridges is removed and set aside for use in preparing for subsequent tests. Prior to this, the tube connecting the tester to the test apparatus is disconnected at a point between the apparatus and the flow-regulating screw clamp, the setting of which is not disturbed. The flow of carbon tetrachloride is started in the test apparatus using the bypass. The test cartridge or cartridges are placed in the apparatus and the test made as usual, the tester being connected to the test apparatus immediately after the test has been started. A leakage of approximately 5 ppm. of carbon tetrachloride vapors is indicated by an unmistakable green coloration of the flame above the copper sleeve. The flame should be observed against a black background in a darkened environment.

20.9 Before each test, the sleeve should be cleaned as follows: The sleeve is held by crucible tongs in the oxidizing portion of a bunsen flame, about 3 inches high, for one minute or until the flame is no longer colored green. This serves to burn off accumulated carbon and chlorinated material. The sleeve is then cooled and totally immersed in 30 cc of a 1 to 1 aqueous solution of concentrated hydrochloric acid to which about one half gram of cupric chloride ($CuCl_2.2H_2O$) has been added, for three minutes or until the entire sleeve has a bright crystalline copper appearance. The sleeve is then rinsed with water and dried on a towel. It is held in the oxidizing portion of a bunsen flame for about a half minute to give it an even coating of black copper oxide and to drive off the residual chlorinated material. After cooling, any loose oxide scale is removed by gentle brushing with a pipe stem cleaner having several coils on the end so that it fits the inside of the sleeve snugly. The sleeve should be handled carefully throughout this cleaning period to avoid deformation. The above cleaning procedure can be readily carried out in a ten minute period. It would be convenient to have several sleeves prepared and brought through the acid and rinsing stage of the cleaning procedure, so that it would only be necessary to heat the sleeve for one half minute and brush it as described above prior to placing it on the test burner.

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